

Project Design Phase-II Technology Stack (Architecture & Stack)

Date	06-7-2025
Team ID	Individual
Project Name	India's Agricultural Crop Production Analysis (1997-2021)
Maximum Marks	4 Marks

Technical Architecture:

The deliverable includes the architectural diagram below and the information as per Table-1 & Table-2.

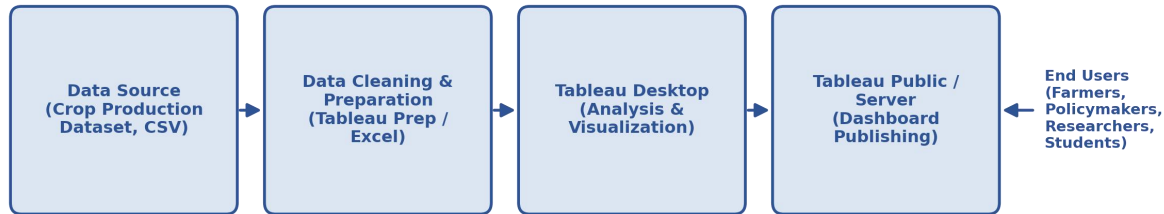


Table-1 : Components & Technologies:

S.No.	Component	Description	Technology
1.	User Interface	Interactive dashboards for exploring crop production trends, yield patterns and state/crop-wise comparisons	Tableau Desktop / Tableau Public
2.	Data Source	India crop production dataset (1997-2021) with state, district, crop, season, area, production and yield	CSV / Excel (data.gov.in / Kaggle - India Agriculture Crop Production Dataset)
3.	Data Cleaning & Preparation	Handling missing values, standardizing state/crop/season names, correcting units and duplicate records	Tableau Prep Builder / MS Excel
4.	Data Storage	Storage of raw and cleaned dataset files used as the data source	Local Filesystem / Excel Workbook
5.	Data Visualization Engine	Building trend charts, maps, comparisons and calculated fields for year-wise, state-wise and crop-wise production analysis	Tableau Desktop
6.	Dashboard Publishing	Hosting and sharing interactive dashboards with end users	Tableau Public / Tableau Server
7.	Report Generation	Exporting summary insights and visuals for offline sharing	Tableau (Export as PDF / Image)

S.No.	Component	Description	Technology
8.	Infrastructure (Server / Cloud)	Application accessed locally during development and published to the cloud for sharing	Local System, Tableau Public Cloud

Table-2: Application Characteristics:

S.No.	Characteristics	Description	Technology
1.	Open-Source Frameworks	List the open-source frameworks used	None (Tableau Public used as a freemium visualization tool)
2.	Security Implementations	Access controls implemented to protect the crop production dataset and published dashboards	Restricted local file access, Tableau Public/Server viewer permissions
3.	Scalable Architecture	Justify the scalability of the architecture (single Tableau workbook covering 25 years of data)	Tableau Data Extracts, incremental data refresh
4.	Availability	Justify the availability of the dashboard to end users	Tableau Public hosting with minimal downtime
5.	Performance	Design consideration for dashboard load speed with a large historical (1997-2021) dataset	Extracts (instead of live connection), filters, optimized calculated fields